

Section A

Answer **all** questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

1. (a) Use the information in **Table 1** to answer the following questions.
- (i) Europa is an asteroid. Estimate its temperature and orbital period.

[2]

Temperature = °C

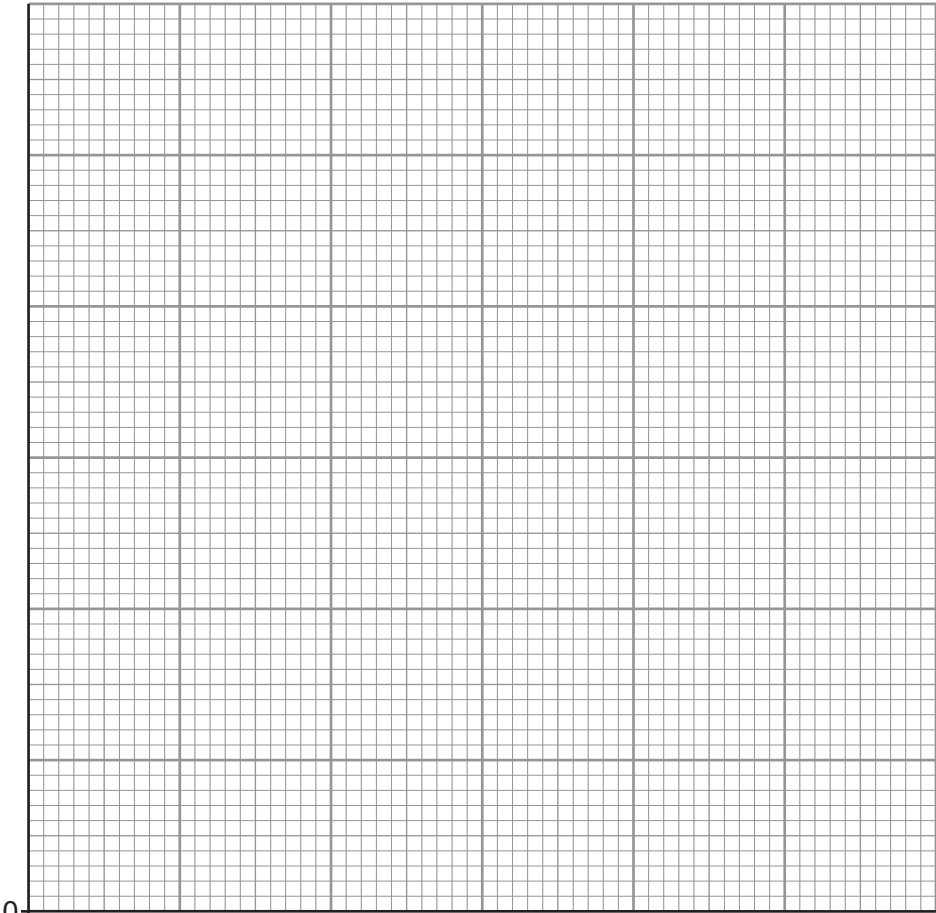
Orbital period = days
- (ii) Pluto is no longer classed as a planet. State **one** reason why.

[1]

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- (iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line.

[4]

Orbital
Velocity
(km/s)



Distance from Sun (AU)

- (iv) Orbital velocity is not proportional to the distance from the Sun.
Explain how your graph shows this to be true. [2]

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- (b) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

- (i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

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- (ii) State **two** ways in which human activities may be affected by solar flares. [2]

1.

2.

- (iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

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- (c) Use the information in **Diagram 1** to describe the differences between Aristotle's model of the Solar System and the 2006 model. [4]

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(d) Use the information in **Diagram 3** to answer the following questions.

(i) Explain what you can deduce about the motion of the three galaxies. [3]

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(ii) Explain what you can deduce about the chemical composition of the three galaxies. [3]

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